

Software Engineer / Web Developer

Jayce Birrell

Armed with a Bachelor of Science in Computing and Software Systems from the University of Melbourne, I've honed my skills in cloud computing, game development, and web app development with 5 years professional experience.

Work History

October 2025 - Ongoing

Entrust ICT, Remote- Contractor

- Designed and implemented RESTful API endpoints for jBilling/NGBilling to support automated customer, product, service, and order creation workflows across complex multi-entity structures
- Built and maintained backend features using Groovy, Grails, Spring, Hibernate, and PostgreSQL, including custom controllers, DTOs, meta-field handling, and error-response structures.

April 2025 - Ongoing

Moodle HQ, Remote - Software Developer

- Developed and maintained core Moodle features across areas including LTI, quiz, assignment, file storage, question bank custom fields, accessibility, and deprecation updates resulting in improved stability and consistency across major releases.
- Delivered high-quality code across multiple Moodle branches, following strict peer-review, testing, and integration processes ensuring compliance with Moodle's strict integration and testing standards.
- Collaborated cross-functionally with developers, QA, product managers, and MoodleCloud infrastructure engineers to plan, troubleshoot, and deliver new platform features.
- Contributed to documentation, upgrade notes, and developer experience improvements to support the wider open-source Moodle community.

October 2021 - September 2024

Entrust ICT, Abbotsford- Web Applications Developer

- Mentored junior developers, providing guidance on best practices, code optimisation, and troubleshooting.
- Managed Linux servers, ensuring uptime, security, and performance optimisation.
- Developed and maintained web applications, integrating APIs and optimising database queries for efficiency.

August 2021 - October 2021

Entrust ICT, Abbotsford- Web Applications Intern

- Designed and developed a product viewer application based on internal requirements.

: <https://www.linkedin.com/in/jayce0808/>

: <https://github.com/Jayce0808>

: <https://jayce.dev/>

References available upon request.

Skills

Software Development & Architecture -

Strong experience designing and implementing scalable backend systems, API integrations, and distributed workflows.

Debugging & Problem Solving – Skilled at diagnosing complex issues across PHP, Groovy/Grails, Postgres, Linux servers, and multi-service environments.

Collaboration & Communication – Effective cross-functional collaboration with developers, QA, PMs, and infrastructure teams in agile and open-source environments.

Testing & Quality – Experienced with PHPUnit, Behat, automated testing, code reviews, and strict integration pipelines.

Mentoring & Leadership – Provided guidance to junior developers and contributed to improving developer experience and documentation.

Adaptability & Learning – Quickly picks up new tools, languages, and large codebases (Moodle's 20+ year codebase, Grails, jBilling, multi-branch workflows).

Technical Skills

Backend / Server

- PHP (Moodle internals), Groovy, Grails, Java, Node.js
- REST API design (enterprise billing APIs, Moodle APIs, CORS, payload schemas)
- PostgreSQL, MySQL, Hibernate ORM
- Multibranch development workflows, version control, code integration

Projects

Staff Accounting Portal

- A **WordPress-based internal portal** designed for staff to manage customer payments efficiently.
- Integrated with **external billing APIs and a CRM system**, allowing seamless import of new sales into the company's billing infrastructure.
- Developed **custom WordPress plugins and hooks** to automate invoice generation and reconciliation.
- Implemented role-based access control to ensure sensitive financial data remained secure.

Interactive Wholesale Website

- A **WordPress-powered e-commerce platform** built with a fully custom theme tailored for wholesale business partners.
- Enabled partners to **sell products to their customers** and track their sales in real-time.
- Implemented a **commission-based system**, allowing partners to claim earnings from their referrals.
- Integrated payment gateways and **custom reporting tools** to streamline order processing and financial tracking.

Integrated Billing System

- A customised script-driven system that processes multiple billing inputs into a standardised invoice format for all business customers.
- Automated the reconciliation of service usage, discounts, and recurring charges from different sources.
- Utilised Python and shell scripting to process large datasets efficiently.
- Reduced manual invoicing errors and improved billing accuracy across departments.

Portfolio

- A **personal portfolio website** built from scratch using **Vue.js**, showcasing professional projects and experience.
- Implemented a **dynamic project showcase**, allowing visitors to filter and explore different works.
- Designed a **minimalist and responsive UI**, ensuring an optimal viewing experience across devices.
- Hosted on a **custom domain with optimised performance**, leveraging lazy loading and caching techniques.

Education

February 2019 - November 2021

University of Melbourne, Parkville—
Bachelor of Science (Computing and Software Systems)

January 2016 - December 2028

Australian Science & Mathematics School, Bedford Park, SA—
SACE Year 12

Frontend

- JavaScript (ES6), Vue.js, React (intermediate), jQuery
- HTML5, CSS3, Semantic UI
- WordPress + custom theme/plugin development

Tooling & Environments

- Linux (Ubuntu, WSL2), Apache, SSL (mkcert, cert provisioning)
- Docker (Grails/Tomcat images, Postgres containers)
- Bash scripting, Python automation

Cloud & DevOps

- Azure fundamentals, CI/CD workflows, environment provisioning
- Familiarity with infrastructure systems (Rainmaker provisioning daemon, MoodleCloud workflows)

Other

- Unity (C#), basic ML/AI, CRM platforms (SugarCRM, SuiteCRM)

Qualifications

October 2023

The Linux Foundation – Certified IT Associate

October 2024

Microsoft – Microsoft Certified: Azure Fundamentals

October 2024

Microsoft – Microsoft Certified: Azure AI Fundamentals